



Problem

Given an array of n integers $x_1, x_2, \dots, x_n$, our task is to find the maximum sub array sum, i.e., For example, in the array $[-1, 2, 4, -3, 5, 2, -5, 2]$ the following subarray produces the maximum sum 10: $[-1, 2, 4, -3, 5, 2, -5, 2]$

Input Format

Two line input First line should read an integer value which represents size of array Second line consists list of elements(integers) separated by space

Constraints

1

Output Format

It prints integer value which represents maximum sub array sum

Sample Input 0

8
-1 2 4 -3 5 2 -5 2

Sample Output 0

10

Sample Input 1

7
-3 4 -1 2 1 -4 3

Sample Output 1

6

Sample Input 2

8
-1 2 4 -3 5 2 -5 2

Sample Output 2

10

Sample Input 3

6
-1 2 3 4 5 -15

Sample Output 3

14

Change Theme

Language

Python 3



```
1 # Enter your code here. Read input from STDIN.
2 def max_subarray_sum(arr:list[int])>int:
3     max_sum=arr[0]
4     end=0
5     for i in range(len(arr)):
6         end+=arr[i]
7         if max_sum<end:
8             max_sum=end
9         if end<0:
10            end=0
11    return max_sum
12 n=int(input())
13 arr=list(map(int,input().split()))
14 print(max_subarray_sum(arr))
```

Line: 12 Col: 14

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0

Compiler Message



Test case 1

Success



Test case 2



Test case 3



Test case 4



Test case 5

